Genetic programming for swarm behaviour optimisation

Ulysse Bourdic-Girard
Ecole Nationale de l'Aviation Civile
Toulouse, France
ulysse.bourdic-girard@alumni.enac.fr

Guillaume Claudon
Ecole Nationale de l'Aviation Civile
Toulouse, France
guillaume.claudon@alumni.enac.fr

***Abstract*—This study presents a method of leveraging genetic programming principles to determine the flight parameters of a drone swarm controlled by a bio-inspired algorithm developed by Matthieu Verdoucq. The study explores how this method can be used to optimise the behaviour of a drone swarm in a search-and-rescue scenario.**

***Index Terms*—Swarm control, Genetic programming**

I. Introduction

Single unmanned aerial vehicles (UAVs) have proven to be exceptionally effective at a wide range of tasks, ranging from cinematography to pesticide application. Swarms of UAVs, although not yet commonly used, hold great promise for future applications. One of these is search and rescue (SAR). In such scenarios a drone swarm has an evident numerical advantage over a single UAV. Controlling the swarm and ensuring it efficiently covers the search area is however a significant challenge.

Our study is based on the bio-inspired flocking algorithm developed by [1] . A swarm controlled by this algorithm can exhibit emergent behaviours such as schooling (the drones flock together in an orderly fashion), swarming (the swarm remains tight-knit but the drones move erratically), or milling (the drones rotate around swarm's centre in an orderly fashion). [2] have demonstrated these behaviours emerge from a specific set of parameters which define the equations used to control the drones.

Our goal is to implement genetic programming paradigms to optimise this set of parameters to adapt the swarm's behaviour to a SAR scenario.

II. Related Work

The work presented in this article is a continuation of the research done by Matthieu Verdoucq for his thesis, which he successfully defended in February 2024.

A. Flocking algorithm

The flocking algorithm developed by [1] is inspired by the behaviour of a school of Rummy-nose tetra (*H. Rhodostomus*). [3] were able to extract a model of social interactions between the fish which leads to emergent behaviour. Matthieu Verdoucq then adapted this model to the dynamics of drones by adding, amongst other things, a vertical component to the social interactions.

[4] have shown that Rummy-nose tetra only interact with a limited number of their neighbours. Matthieu Verdoucq implemented this in his interaction model which means the drones need only gather a limited amount of information from their social environment to display coordinated behaviour.

The emergent behaviour of the swarm is influenced by a set of parameters which determine the guidance of each individual drone in the swarm. [5] have shown how changing these parameters can alter the behaviour of the swarm in real-time.

These studies have observed that certain combinations of parameters induce the same behaviour in the swarm. Each of these specific behaviours is called a phase. The research produced by Matthieu Verdoucq for his thesis principally focuses on phases emerging from interactions between the alignment and attraction coefficients γ_{ali} and γ_{att}. [1] has identified three distinct phases linked to these parameters: schooling, swarming and milling.

This means that although the paths of individual UAVs in the swarm cannot be controlled, their global behaviour can. However, there are many other parameters which influence the behaviour of the swarm which were not studied by Matthieu Verdoucq. The complexity of the interactions between these parameters prevents a manual tuning of this set of parameters. Genetic programming paradigms have been used to optimise drone swarm objectives and are promising means of accomplishing our objective: optimising the set of drone guidance parameters to a specific task.

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